

K2-30b

R-2373

Benchmark run · workflow W8-benchmark-deep · record deep-bench_K2-30_210113 · created 2026-08-20T00:34:11+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459227.8183 +/- 0.0081 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.129 +/- 0.04
Transit depth (Rp/Rs)^2	1.66 +/- 1.03 [%]
Orbital Inclination (inc)	89.92 +/- 2.15
Airmass coefficient 1 (a1)	0.961 +/- 0.025
Airmass coefficient 2 (a2)	0.0145 +/- 0.0032
Scatter in the residuals of the lightcurve fit is	2.63 %
Best Comparison Star	#2 - [467, 343]
Optimal Aperture	3.58
Optimal Annulus	13.16
Transit Duration (day)	0.102 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.129 ± 0.04 vs 0.1266 (deviation 0.06σ)
Duration fit vs published	0.102 d vs 0.0971 d (deviation 0.31σ)
Mid-time O–C	21.8 min
Depth SNR	3.2
Residual scatter	2.63 %

Light curve

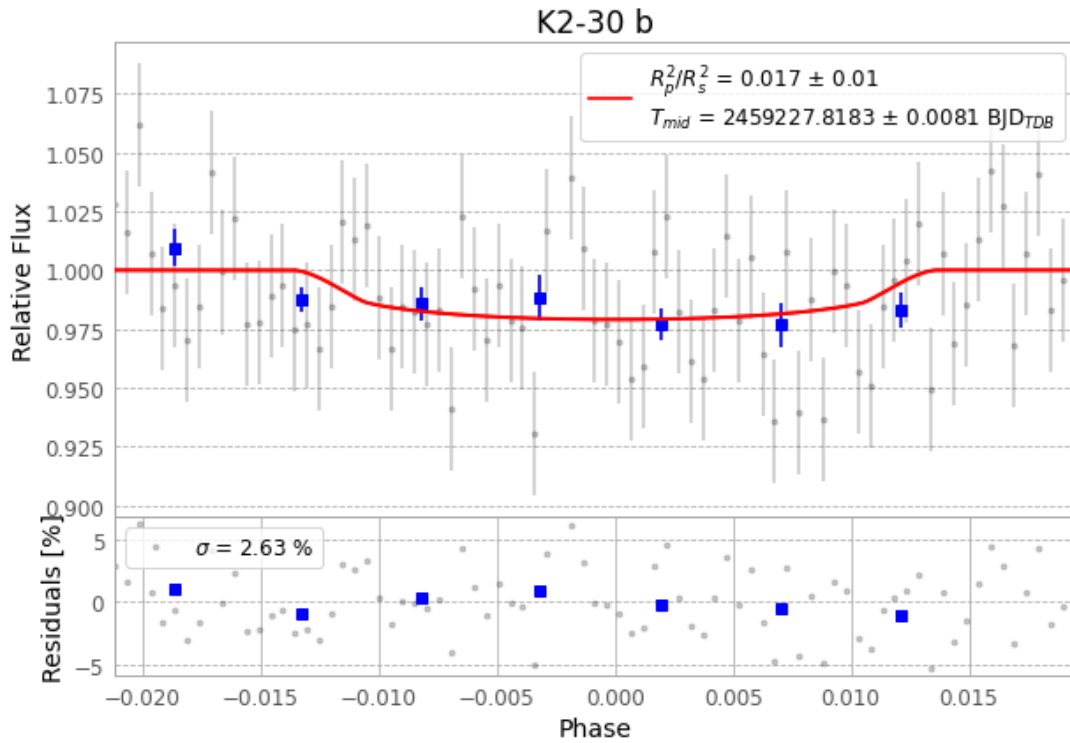


Figure 1 — FinalLightCurve_K2-30 b_2021-01-12.png (SHA-256 9dcd1b3ca8e3eea4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [366, 268], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 1856d5ce2f349cee...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

81 FITS frames (53 MB), K2-30210113052714.FITS → K2-30210113092712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-K2-30-210113.log (9246231ea7d8...), FinalLightCurve_K2-30 b_2021-01-12.png (9dcd1b3ca8e3...), FinalLightCurve_K2-30 b_2021-01-12.csv (8482d8c6f435...)

Compute resources

Wall time 108.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-20 00:34Z.